- United States Government - *http://www.data.gov/*
- Machine Learning - *http://bitly.com/bundles/bigmlcom/2*
- Scientific Data University of Muenster - *http://data.uni-muenster.de/*
- Hilary Mason research-quality - *https://bitly.com/bundles/hmason/1*

The other way to get data sets is to generate our own data sets using programming languages. The major issue here is the selection of a suitable programming language or tool that can fetch live data from social media applications or live streaming portals. The usual programming languages are not effective in fetching live streaming data from the Internet.

Python is one of the outstanding and efficient programming languages that can communicate with the live streaming servers. You can use it to store the fetched data in the database or file system for analysis and predictions.

Let's move on to look at some real-life cases in which Python has been used to fetch live streaming data.

Web scraping using Python scripts

Python scripts can be used to fetch live data from the Sensex. This technique is known as Web scraping.

Figure 1 gives a screenshot of the live stock market index from *timesofindia.com*. The frequently changing Sensex is fetched using Python and stored in a separate file so that the record of every moment can be saved. To implement this, the library of *BeautifulSoup* is integrated with Python.

The following code can be used and executed on Python. After execution, a new file named *bseindex.out* will be created and second-by-second Sensex data will be stored in it. Once we flood the live data in a file, this data can be easily analysed using data mining algorithms with SciLab, WEKA, R, TANAGRA or any other data mining tool.

```
from bs4 import BeautifulSoup
import urllib.request
from time import sleep
from datetime import datetime
def getnews():
    url = "http://timesofindia.indiatimes.com/business"
    req = urllib.request.urlopen(url)
    page = req.read()
    scraping = BeautifulSoup(page)
    price = scraping.findAll("span",attrs={"class":"red14px"})
[0].text
    return price
with open("bseindex.out","w") as f:
    for x in range(2,100):
        sNow = datetime.now().strftime("%I:%M:%S%p")
        f.write("{0}, {1} \n ".format(sNow, getnews()))
        sleep(1)
```

Using a similar approach, YouTube likes can be fetched and analysed using Python code, as follows:

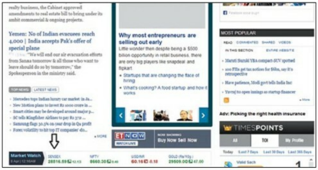

Figure 1: Screenshot of *timesofindia.com* to analyse Sensex attributes

Figure 2: A screenshot of the *Baahubali* movie trailer from YouTube

```
from bs4 import BeautifulSoup
import urllib.request
from time import sleep
from datetime import datetime
def getnews():
    url = " https://www.youtube.com/watch?v=VdafjyFK3ko "
    req = urllib.request.urlopen(url)
    page = req.read()
    scraping = BeautifulSoup(page)
    price = scraping.findAll("div",attrs={"class":"watch-view-
count"})[0].text
    return price
with open("baahubali.out","w") as f:
    for x in range(2,100):
        sNow = datetime.now().strftime("%I:%M:%S%p")
        f.write("{0}, {1} \n ".format(sNow, getnews()))
        sleep(1)
```

Fetching all the hyperlinks of a website

All the hyperlinks of a website can be fetched by using the following code:

```
from bs4 import BeautifulSoup
import requests
newurl = input ("Input URL")
record  = requests.get("http://" +newurl)
```